effects for subjects and items, by running SBC. For the bridge sampling, we again used the method "warp3" with the aim to keep the number of warning messages of the bridge sampling algorithm low. We first investigated model convergence. Across all model fits (200 simulations $\times 2$ models (H0/H1) = 400 fits), we observed 0 divergent transitions. R-hat exceeded the threshold of 1.01 for at least one population-level parameter in 0 cases, suggesting good model convergence. During the bridge sampling, there was the warning message "logml could not be estimated within maxiter, rerunning with adjusted starting value. Estimate might be more variable than usual" in 3 of simulations, suggesting only few problems with the bridge sampling. Therefore, we performed statistical analysis only for simulations without a warning message from the bridge sampler.

The results (see Fig. 2) showed that the average posterior model probabilities for the H1 did not differ from the prior (marginal SBC). Indeed, default Bayes factor analyses with Bayes factors of $BF01 > 10$ provided support for the null hypothesis that the average posterior probability for H1 did not differ from the prior of 0.2, and sensitivity analyses showed that this result was stable across a larger range of prior scales. The results from the reliability diagram (see Fig. 2C) also did not provide evidence for bias, but rather suggested accurate Bayes factor estimation when no warning message was shown.

**Design 3: 2x2 repeated measures design with crossed random effects for subjects and items**

We studied a repeated measures $2 \times 2$ Latin square design with crossed random effects for subjects and items. We again used the "warp3" method for bridge sampling, and used $10,000$ iterations of MCMC sampling. We first investigated model convergence. Across all model fits (200 simulations $\times 3$ effects $\times 2$ models (H0/H1) = 1200 fits), we observed 0 divergent transitions. R-hat exceeded the threshold of 1.05 for at least one population-level parameter in 1 case, suggesting overall good model convergence. However, during the bridge sampling, there was the warning message "logml could not be estimated within maxiter,

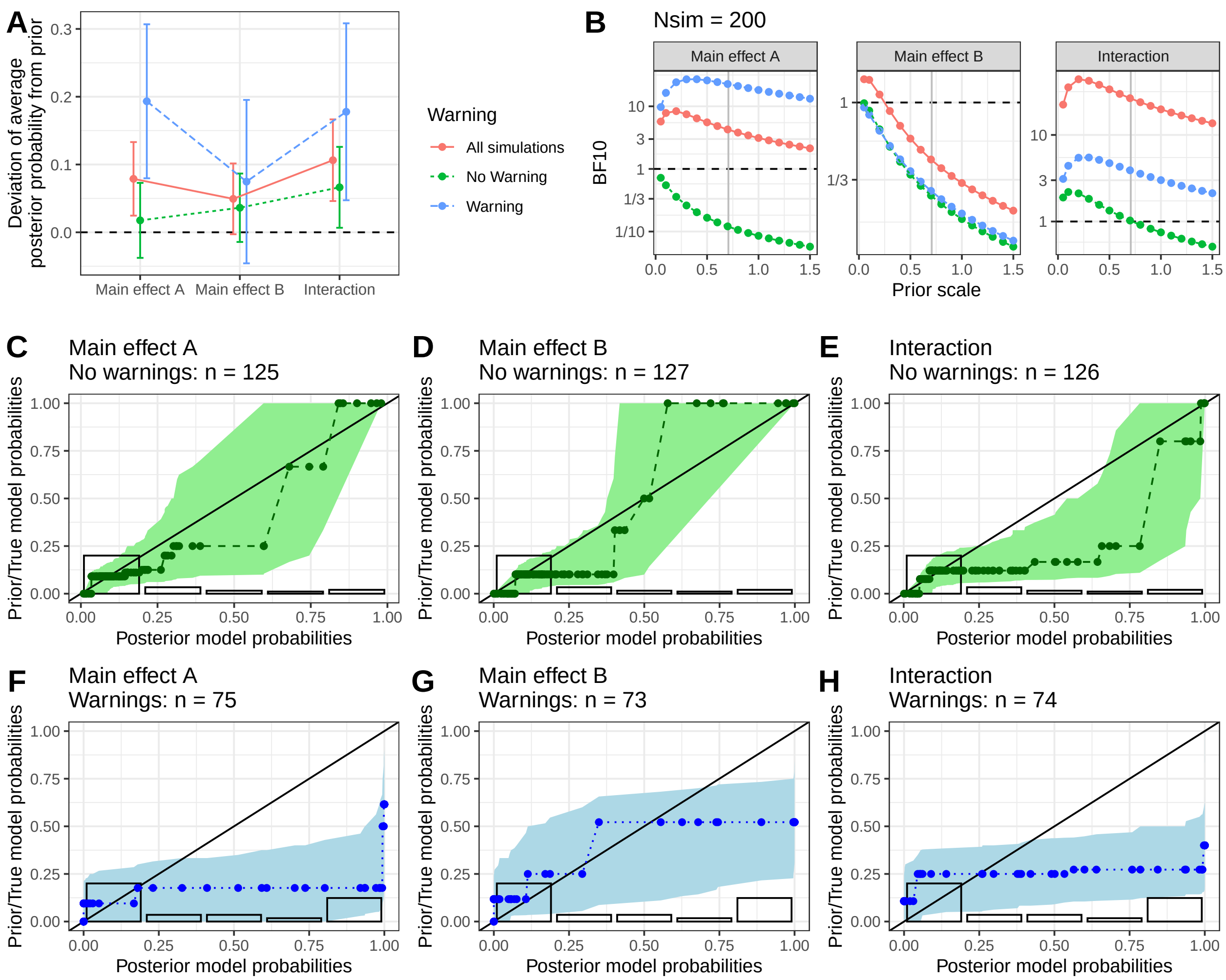


*Figure 3*. A 2 x 2 repeated measures design with crossed random effects for subjects and items. A) The average deviation of the posterior model probability from the prior (95% CI) is shown for the main effects A, B, and their interaction. Data is shown for all simulations and separately for those without / with a warning message from the bridge sampling algorithm. The horizontal broken line indicates not deviation; deviations from this line indicate bias. B) Bayesian t-tests of whether the posterior probability for H1 deviates from the prior, with sensitivity analyses for the prior scale. The vertical grey line indicates a default Bayes factor. C-H) Reliability diagrams: the prior model probability is plotted as a function of the estimated posterior model probabilities for simulations without (C-E) and with (F-H) a warning message. Error bands show 95% consistency bands. Results show that when no warning is issued, prior model probabilities largely do not deviate from posterior model probabilities. Results for the interaction are unclear. However, biases are visible in simulations with a warning message.

rerunning with adjusted starting value. Estimate might be more variable than usual" in 37% of simulations, suggesting problems with the bridge sampling. We again looked at the results separately for simulations with versus without warning messages from the bridge sampler.

The results (see Fig. 3) showed that – for simulations without warning messages from the bridge sampling – the posterior model probabilities did not differ from the prior model probability for the main effects. Indeed, the null hypothesis that the posterior model probability was equal to the prior was supported by Bayesian t-tests (see Fig. 3B) showing Bayes factors of $BF01 > 3$ for the null hypothesis for the two main effects (sensitivity analyses showed stronger effects for main effect A than B), and reliability diagrams did not show clear evidence for bias (Fig. 3C-D). For the interaction, the results were not quite clear: the results from the Bayesian t-test strongly depended on the prior scale; moreover, a slight deviation seemed present in the reliability diagrams (see Fig. 3E) and more SBC simulations may be needed to judge whether this deviation is reliable.

For simulations with a warning message, the results from the marginal SBC were mixed, with evidence for bias ($BF10 > 3$) in the case of main effect A and the interaction, but slight evidence for absence of bias in the case of main effect B ($BF01 > 3$). However, reliability diagrams (see Fig. 3F-H) indicated rather clear evidence for miscalibration between posterior and prior probabilities, suggesting Bayes factor estimates were biased.

These results suggest evidence for absence of overall bias in the Bayes factor estimates for the main effects when no warning message is issued. The results for the interaction were unclear. However, in the cases where a warning message was issued by the bridge sampler, we saw evidence for bias in the Bayes factor estimates, with posterior model probabilities (for main effect A and for the interaction) being too large and larger than the prior probability (at least for large posterior model probabilities).

**Increasing the number of MCMC iterations to 20,000.** The previous simulations suffered from a relatively large number of warning messages of the bridge